International Health 3 (2011) 35–43

Contents lists available at ScienceDirect

International Health

journal homepage: http://www.elsevier.com/locate/inhe

Dowry deaths: a neglected public health issue in India

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ARTICLE INFO

Article history:
Received 3 February 2010
Received in revised form 2 October 2010
Accepted 15 December 2010
Available online 12 February 2011

Keywords:
Dowry
homicides
suicides
domestic violence
public health
India

ABSTRACT

This paper appraises the public health burden of mortality in India caused by the practice of dowry and examines the association of some demographic and socio-economic factors with dowry deaths and dowry-related suicides. The paper is based on the data available on the public domains of the National Crime Records Bureau (NCRB), third National Family Health and Survey-2005-06, Planning Commission of India and Census of India 2001. In 2007, the total number of dowry deaths and dowry-related suicides reported in India were 8093 and 3148, respectively. There was a 74% increase in dowry-related deaths from1995 to 2007, while there was a 31% increase in the reporting of dowry-related suicides. Occurrence of dowry deaths has significant association with some demographic and socio-economic variables. The data reveal that the status of women is undesirable, and the burden of mortality and related morbidity is enormous. There should be a national injury surveillance system and reliable estimates of dowry-related homicides. However, available information can be used to design and implement some counter-measures to prevent dowry-related violence and deaths. The study warrants the undertaking of research to give insights into circumstances and triggers of such violence, the healthcare seeking of these victims, bottlenecks in seeking health care and reporting to the police.

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1. Introduction

Dowry has often been referred to either in a narrative or tale, but in the past, it might not have the prevailing negative connotations as a social evil. The purpose of this practice was meant to help new couples start their life in comfort. The dowry system, giving cash and kind from the bride's family to the groom or groom's family at the time of marriage, has a long history in India and other Asian societies.[1] The practice of dowry was not uncommon in Europe, East Asia, South Asia, and pockets of the Americas. Though the custom has largely disappeared in the western world, it remains in South Asia.[2] One feature of the dowry system in general, which applies specifically to India, is its association with socio-economic stratification. Dowry helps in perpetrating these divisions over generations.[3] Rao identified some fundamental patriarchal conditions that lie at the root of the problem in India;[4] they include: (i) a woman's primary role as a mother and daughter with limited options outside marriage, (ii) a daughter who stays unmarried beyond a certain age whose parents face high social costs, (iii) divorce is practically non-existent, (iv) women are customarily denied inheritance rights and (v) marriages are arranged by the parents of brides and grooms to largely reflect the interests of parents. Though India possesses several communities which are discrete in their geography, language and culture, marriage is perceived as a universal obligation, customarily arranged by parents or relatives of the bride and groom[5] and dowry became a major constituent in marriage negotiations. Now the money and property, as a tangible part of marriage settlements, has been accepted as a private family matter. It has

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doi:10.1016/j.inhe.2010.12.002

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become more private and confined to both families, due to the legal implications of giving and accepting dowry. Most of the time, its unfavourable and disastrous consequences are superficially weighed as individual destiny.

In a patriarchal society, women have little or no role in the market economy and are dependent upon their husbands and in-laws.[6] Today the practice of dowry, which was once limited among some communities, has become customary across caste and class groups in India due to changed economy.[7] Modernization has increased the desirability of consumer goods, and young married couples may see dowry as a way of obtaining them. Studies show that even parents of daughters see dowry as an opportunity to make status claims and to ensure good treatment of their daughters by their conjugal families.[8] Traditionally, dowry items mainly consisted of clothing and household items, but have now been replaced by cash or large consumer goods due to competition for 'a better groom'.[5] Social scientists provided two major explanations of this shift, one demographic and the other status based.[4] The demographic explanation lays the blame on population growth; the other explanation is the drive towards social status as mentioned above. The practice of dowry has now become a depraved and an ignoble commercial transaction in which financial considerations receive preference over all other qualities of the bride. Even though the custom of dowry has become a source of serious threats to the life of women in many cases, paradoxically it receives support not only from society but also from women themselves.[7] The practice of giving and taking dowry was legally prohibited in 1961, but still continues in 2010.

1.1. Dowry deaths – a public health issue

Though the marriage deal is completed by paying dowry, there is no guarantee for the health and well-being of the bride after marriage. There are many instances reported in the public media of the groom or his family making another demand for dowry after marriage. This is usually due to lower-value dowries being given than the offer made during the deal or may be due to the influence of patriarchal factors to demand more dowries from the bride's family. Women whose families paid either fewer or lower-value dowries, those whose in-laws have expressed dissatisfaction with their dowries, and those who have faced post-marriage dowry requests have been repeatedly found to be more likely to report domestic violence against them.[9–12] The issue is not limited to number of deaths only. The magnitude of the morbidity in terms of deterioration of physical and mental health due to dowry-related conflicts is enormous. The family-level conflicts and associated violence have a significant role in determining many social and health conditions of women as well as children.[13–20] There are cases of poor utilisation of ante-natal care and child immunisation due to the inferior consideration of the women and conflicts within the family.[21] The issues of female foeticide, high infant mortality, maternal mortality, malnutrition of women and female children should also be read synchronically. Despite the enormous impact of dowries on women's health, little information is available in public health literature. In this context, the present paper examines the trend of the mortality in the form of dowry deaths and dowry-related suicides from India. The association of various demographic and socio-economic factors with these events are also examined.

1.2. Conceptual framework

Violence against women is often examined from an ecological perspective to understand its risk factors.[22,23] This ecological framework conceptualizes dowry-related deaths and other violence against women as multi-faceted phenomena. Heise noted that this phenomenon is grounded in an interplay of individual, family and community-level factors.[23] The present study too uses this framework to examine the risk factors of dowry-related deaths.

Research suggests that some women are more likely to be victimized than others, because of individual differences in their abilities, resources and experiences. Literacy, for example, has an impact on coping with family conflicts and lessens the risk of victimization due to these conflicts. Education attainment in women improves access to knowledge and information, and brings about an openness to ideas and an increasing independence from traditional authority.[24] Age at marriage and teenage pregnancy also have an impact on the victimization of women. Early age at marriage jeopardizes women's status and is likely to make motherhood the sole focus of her life. It limits women's opportunities in education, occupational training and employment. Further, a lack of education and employment undermines the status of women in the family and community. Also, research revealed that the younger the woman marries, the more likely she is to experience marital conflicts.[25] Women's exposure to mass media makes them aware of their opportunities and rights with the result that they are less likely to experience dowry-related violence.

Some characteristics of couples and families influence a woman's likelihood of being victimized. Family-level characteristics include economic and decision making power and the role of women in the family. Family-level stressors like husband's unemployment and poverty also increase a woman's likelihood of being victimized. A husband's inability and lack of resources to meet the family's needs can cause stress and ultimately lead to frustration. Violence against women is a possible response to this frustration. There is the possibility of increase of alcohol abuse among these men and alcohol consumption among men is positively associated with the perpetration of violence against women. Also, the power dynamics in the family and women's empowerment have an impact on women's status and likelihood of victimization. This likelihood is lower among women who are involved in the family's decision making process. Similarly women's involvement in income-generation activities (like microfinance schemes) helps them, to an extent, to access knowledge and information and attain power to take part in decision making.

Communities have inherent features of protecting people from violence and conversely making people vulnerable to victimization in different contexts. Some community-level characteristics are a reflection of the lower status of women in the community; lower sex ratio is one such kind.

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Similarly rates of crimes and, specifically, those against women indicate the level of integrity of the community and to what extent the safety of women is assured by the community. The brunt of violence in the community is borne by women too.

2. Methods

2.1. Data

Data of both outcome variables and independent variables were taken from the public domains of governmental and related organizations. Details of these data, including the source are presented herewith. India is a republic, consisting of 28 states and seven union territories with a democratic parliamentary system. Only 15 states (Andhra Pradesh, Assam, Bihar, Gujarat, Haryana, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Orissa, Punjab, Rajasthan, Uttar Pradesh, West Bengal, and Tamil Nadu), which possessed all the data have been considered for this analysis. These states are in the North, South, Central, East, West and North-Eastern zones of India.

Data of the outcome variables were taken from the public domain of the National Crime Records Bureau (NCRB). The NCRB, an organization under the government of India, functions as a clearing house for information related to crimes occurring in the country. Through Crime Criminal Information System (CCIS), NCRB developed a national network, with the police headquarters at 727 police districts and 35 state/union territory levels connected to one another, and to the central data repository at the NCRB in New Delhi, using standard input integrated investigation forms. The application resides at the NCRB office in New Delhi, and runs in district headquarters across the country. Records fed in at the district level are periodically updated at the centre through the offline/online system of data transfer. The integrated information forms serve as the backbone of the application, which links 140 tables, 120 forms and 160 parameters related to crimes, criminals and property. Thus, this network amassed data from 15 000 police stations through district and state/union territory levels for national figures at national level at the central data repository at the NCRB in New Delhi.[26]

Data of the independent variables were taken from the third National Family Health Survey 2005–06,[27] the Planning Commission of India[28] and the Census of India 2001.[29] State-level values of these variables are available on their public domains as well as in the form of reports.

2.2. Outcome variables

Outcome variables considered for this paper are dowry deaths and suicide where dowry is a cause. Dowry death means the death of a married woman who was murdered by her husband and/or in-laws. Suicide-related dowry is a suicide in which dowry is the principal cause. Dowry-related suicides are included while counting the dowry deaths. These deaths have been recorded based on the first information report recorded by the police at community level. However, it is not known how many of these cases were proved in courts of justice.

2.3. Independent variables

Independent variables are considered based on the ecological framework mentioned under conceptual framework. The variables at three levels were included for the analysis. They are: individual-level variables of women (age at marriage, teenage pregnancy, literacy, unemployment and exposure to media), family-level variables (unemployment of men, per capita income, spousal violence, women's role in decision making in the family, women's access to microfinance schemes and men's alcohol habit) and community-level variables (sex ratio, crime rate and rate of crimes against women).

Age at marriage is the percentage of women aged 18–29 years who were first married by the age of 18. Teenage pregnancy is the percentage of women who have begun childbearing during the ages 15–19 years. Literacy rate is the percentage of women and men, aged 15–49 years, who can read a whole sentence or part of a sentence and women and men who completed Standard 6 or higher. Unemployment is the percentage of women and men aged 15-49 years who are not employed in the 12 months preceding the survey.[29] Exposure to mass media is the percentage of women aged 15–49 years who are not regularly exposed to any of the media.[27] Per capita income is the average income per individual.[28] Prevalence of spousal violence is the percentage of women aged 15–49 years, who have experienced physical domestic violence.[27] Women's decision making in the family is the percentage of currently married women who usually make four specific kinds of decisions (own health care, major household purchases, purchases for daily household needs and visits to her family or relatives) either by themselves or jointly with their husband.[27] Access to microfinance is the percentage of women who have membership of and/or have taken a loan from a microcredit scheme.[27] Men's alcohol use is the percentage of men aged 15–49 years who drink alcohol.[27] Sex ratio is the total number of women for 1000 male population in the State as per Census of India, 2001.[29] Crime rate is the percentage contribution of total crime to all of India in 2007.[26] Crime against women is the percentage contribution of crime against women to all in India in 2007.[26]

2.4. Analysis

Outcome variables (dowry deaths and dowry-related suicides) and other independent variables of fifteen states were compiled. Univariate and multivariate regression analyses were conducted to examine the association of the independent variables. Dependent variables were two outcome variables: dowry deaths and dowry-related suicides. Separate regression analyses were carried out for each dependent variable. Initially, each independent variable was regressed against each dependent variable. Those variables with a minimum P value of 0.25 were considered for multiple regression analyses. Hosmer and Lemeshow recommended a P value of < 0.25 to be used as a screening criterion for variable selection.[30] Multiple regression

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Table 1
Occurrence of dowry deaths and dowry-related suicides in India from 1995–2007.

Year	Dowry deaths	% Variation over the year	Dowry-related suicides	% Variation over the year
1995	4648	–	2407	–
1996	5513	18.6	2180	−9.4
1997	6006	9.0	2496	14.5
1998	6975	16.1	2517	0.8
1999	6699	−4.0	2572	2.2
2000	6995	4.4	2446	−4.8
2001	6851	−2.0	2414	−1.3
2002	6822	−0.4	2410	−0.2
2003	6208	−9.0	2447	1.5
2004	7026	13.2	2638	7.8
2005	6787	−3.4	2351	−10.8
2006	7618	12.2	2336	−0.6
2007	8093	6.2	3148	34.7

Source: National Crime Records Bureau[26].

models were run by stepwise procedure with a P value of >0.10 to exit. All analyses were done using SPSS 17.0 (SPSS Inc, Chicago, USA).

3. Results

In 2007, the total number of dowry deaths and dowry-related suicides reported by NCRB in India were 8093 and 3148, respectively. Table 1 reveals the trend of year-wise occurrence of dowry deaths during 1995-2007 in India. There was a 74% increase in the occurrence of dowry deaths from 1995 to 2007, and there was a 31% increase in the occurrence of dowry-related suicides during this period. Table 2 shows the values of both outcome and independent variables of 15 Indian states. Dowry deaths and dowry-related suicides are shown in the form of number of cases per million population. Of all states, the highest occurrence of dowry deaths and dowry-related suicides are reported from Bihar (followed by Orissa, Uttar Pradesh and Madhya Pradesh), and West Bengal (followed by Madhya Pradesh, Maharashtra and Haryana), respectively.

The univariate regression revealed that occurrence of dowry deaths has significant association with three individual-level variables namely, age at marriage, literacy rate and women's exposure to mass media, and with two family-level variables namely, per capita income and spousal violence (Table 3). With respect to dowry-related suicides, no significant relation is established with any of the independent variables. Of the 14 independent variables, nine variables which met the required criterion were included in multiple regression analysis (Table 4). However, only women's literacy rate is significantly and inversely associated with occurrence of dowry deaths. Similar multivariate regression was carried out with occurrence of dowry-related suicides by considering four independent variables, however, none of them yielded significant association.

The analysis revealed that female literacy has a significant relationship with dowry deaths. From the selected fifteen states, the percentage of women's literacy rate is very low in Bihar (37%), Madhya Pradesh (44%), Uttar Pradesh (45%) and Orissa (52%), where more dowry deaths are reported. In univariate analysis, significant relationships with women's exposure to media and dowry death are found ($P = 0.004$). When there are higher proportions of women without exposure to media, the occurrence of dowry deaths are higher. In Bihar, more than 58% of women have no regular exposure to any kind of media and the total number of dowry deaths in 2007 was 1172. A similar situation is noted in Madhya Pradesh and Uttar Pradesh. Per capita income is another variable found significant during univariate analysis on the occurrence of dowry deaths ($P = 0.017$). Data shows that where the per capita income is low dowry death has been reported high. Physical domestic violence is another variable found significant in univariate regression with dowry-related deaths ($P = 0.029$). More dowry deaths have been reported in the states of Bihar, Madhya Pradesh and Uttar Pradesh than in other states where the prevalence of domestic violence is low. In these states 39%, 37% and 30.3% of women, respectively, have experienced domestic violence.

4. Discussion

The present analysis reveals the increase of both dowry-related deaths and suicides in India during 1995–2007. There is no doubt that many dowry death cases are not reported and so the actual deaths and atrocities due to dowry are more than the present figures show. There are approximately 15 000 dowry deaths estimated per year,[31] which are mostly in kitchen fires designed to look like accidents.[32] Since India does not have a national injury surveillance system and the official records of these incidents are low, the only source of information about the incidence of burn morbidity and mortality is from police reports.[33] Sanghvi et al.[34] reported that 65% of fire-related deaths among women occurred in those aged 15–34 years and were likely to be caused by: kitchen accidents related to use of kerosene and flammability of garments, self-immolation or suicides and homicides related to domestic violence. Another study based on autopsy data of 5933 unnatural deaths over 25 years revealed that females (61%) outnumbered males (39%) in deaths for which cause of death was burns.[35] This autopsy study also revealed that 55.4% of the total unnatural deaths of women and only 12.8% of unnatural deaths in men occurred due

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Table 2
Distribution of dowry deaths, dowry-related suicides and some socio-economic and demographic variables in the selected fifteen states of India.

Variables	Andhra Pradesh	Assam	Bihar	Gujarat	Haryana	Karna-taka	Kerala	Madhya Pradesh	Maha-rashtra	Orissa	Punjab	Rajasthan	Tamil Nadu	Uttar Pradesh	West Bengal
Outcome variables															
Dowry deaths (per million population)	8.04	3.75	14.13	0.83	12.72	4.74	0.84	12.29	4.5	12.52	5.46	7.76	3.33	12.49	5.62
Dowry-related suicide (per million population)	2.66	0.26	0.22	0.92	2.79	0.79	0.69	9.74	3.53	1.92	0.24	0.67	1.55	2.25	12.46
Independent variables															
Individual-level variables															
Age at marriage (% of women who married by the age of 18 years)	56.2	38.2	63.7	35.40	41.4	41.0	17.2	52.60	40.2	37.50	21.6	58.40	25.2	52.2	53.3
Teenage pregnancy (% of women aged 15–19 years who began childbearing)	18.1	16.4	25.0	12.7	12.1	17.0	5.8	13.6	13.8	14.4	5.5	16.0	7.7	14.3	25.3
Literacy rate of women (%)	49.6	63.0	37.0	63.8	60.4	59.7	93.0	44.4	70.3	52.2	68.7	36.2	69.4	44.8	58.8
Unemployed women (%)	47.8	70.2	66.0	47.0	71.7	53.7	69.7	46.0	51.7	63.1	75.2	44.6	50.2	65.8	64.9
Exposure of women to mass media (%)	18.1	38.6	58.2	28.4	32.3	22.1	9.5	46.9	23.6	38.8	15.7	53.1	11.2	47.5	36.0
Family-level variables															
Unemployed men (%)	11.7	15.9	16.3	7.5	15.1	9.6	13.2	12.8	13.4	10.4	13.6	14.9	12.6	14.1	10.0
Per capita income (in Indian Rupees[a])	21 227	14 950	6610	26 543	29 887	21 829	25 657	12 566	29 085	13 329	28 872	15 219	24 038	10 637	20 485
Spousal violence (% of women who experienced physical spousal violence)	29.9	24.7	38.9	20.7	23.4	16.7	12.6	37.0	27.2	24.5	25.0	27.5	36.1	30.3	19.9
Women's decision (% of women who are involved in decision making)	40.4	60.9	32.7	36.6	41.7	35.2	47.2	29.4	45.4	41.8	37.0	22.8	48.8	33.7	23.9
Microfinance (% of women who accessed microfinance schemes)	16.3	1.4	1.0	1.7	0.6	9.2	8.0	1.0	2.1	6.9	0.9	0.6	13.4	0.4	2.7
Alcohol use (% of men who drink alcohol)	47.2	37.8	34.9	16.0	27.7	28.3	45.2	30.8	24.0	39.6	43.4	19.1	41.5	25.3	34.0
Community-level variables															
Sex ratio (number of women per 1000 men)	978	932	921	921	861	964	1058	920	922	972	874	922	986	898	934
Crime (% contribution to all India total)	8.8	2.3	5.5	6.2	2.6	6.1	5.5	10.2	9.8	2.8	1.8	7.5	8.7	7.6	4.1
Crime against women (% contribution to all India total)	13.3	3.7	4.1	4.5	2.5	3.5	4.2	8.3	8.1	3.9	1.5	7.7	4.2	11.3	8.9

[a] 1 Indian Rupee is approximately equivalent to US$0.02.
Source: National Crime Records Bureau,[26] Third National Family Health Survey,[27] Planning Commission of India[28] and Census of India[29].

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Table 3
Details of univariate regression analysis for association of dowry deaths and dowry-related suicides with some socio-economic and demographic variables.

Independent Variables	Dowry deaths		Dowry-related suicides	
	β Coefficient	*P*-value	β Coefficient	*P*-value
Individual-level variables				
Age at marriage (% of women who married by the age of 18 years)	0.631	0.012	0.340	NS
Teenage pregnancy (% of women aged 15–19 years who began childbearing)	0.377	NS	0.381	NS
Literacy rate of women (%)	−0.744	0.001	−0.142	NS
Unemployed - women (%)	0.134	NS	−0.148	NS
Exposure of women to mass media (%)	0.689	0.004	0.177	NS
Family-level variables				
Unemployed - men (%)	0.390	NS	−0.261	NS
Per capita income (in Indian Rupees[a])	−0.604	0.017	−0.075	NS
Spousal violence (% of women who experienced physical spousal violence)	0.563	0.029	0.083	NS
Women's decision (% of women who are involved in decision making)	−0.364	NS	−0.453	NS
Microfinance (% of women who accessed microfinance schemes)	−0.272	NS	−0.129	NS
Alcohol use (% of men who drink alcohol)	−0.052	NS	−0.052	NS
Community-level variables				
Sex ratio (number of women per 1000 men)	−0.440	NS	−0.108	NS
Crime (% contribution to all India total)	−0.012	NS	0.217	NS
Crime against women (% contribution to all India total)	0.197	NS	0.442	NS

[a] 1 Indian Rupee is approximately equivalent to US$0.02; NS = not significant.

to burns. Some micro-studies revealed that the injuries resulting from kitchen accidents,[36,37] self-immolation[38] and domestic violence[36,39] which could include dowry related harassment have lead to death. Also, there were instances in which fire-related homicides are disguised as accidents and suicides.[34]

The number of dowry-related deaths in Kerala is lower than in many other Indian states. The literacy rate among men and women in Kerala is highest among all Indian states which could be one of the reasons for the relatively lower number of dowry-related deaths. Studies show that the dowry system is more rigid in the northern Hindi-speaking region consisting of Bihar, Uttar Pradesh and Rajasthan states.[40] Among the selected fifteen states, the highest number of dowry deaths occurred in Uttar Pradesh followed by Bihar and Madhya Pradesh, which are major Hindi-speaking northern states. Mukherjee et al. made an attempt to understand the regional variations in dowry deaths and other crimes against women.[41] Based on the NCRB data of three years (1995 to 1997), they concluded that the spatial concentration of dowry deaths is in the northern part of India. Age at marriage of women is an important factor with respect to her family life. To avoid giving too much dowry and to get rid of the burden of daughters, parents are marrying off their daughters at the earliest opportunity. As a result the very young bride is not able to resist and cope with the harassment of her husband and in-laws for dowry. The young woman is often helpless and, facing negative experiences of her family life or being victimised by her conjugal family, ultimately thinks that there is no option except committing suicide. Women's exposure to media is an important factor which may empower women to a great extent in their daily life. It is true that women who have more exposure will be

Table 4
Details of multivariate regression analysis for association of dowry deaths and dowry-related suicides with selected socio-economic and demographic variables.

Independent Variables	Dowry deaths		Dowry-related suicides	
	β Coefficient	*P*-value	β Coefficient	*P*-value
Constant	20.515	<0.001	8.630	NS
Individual-level variables				
Age at marriage (% of women who married by the age of 18 years)	−0.125	NS	−0.197	NS
Teenage pregnancy (% of women aged 15–19 years who began childbearing)	−0.147	NS	0.393	NS
Literacy rate of women (%)	−0.228	0.001	NA	NA
Exposure of women to mass media (%)	0.227	NS	NA	NA
Family-level variables				
Unemployed men (%)	0.252	NS	NA	NA
Per capita income (in Indian Rupees[a])	−0.123	NS	NA	NA
Spousal violence (% of women who experienced physical spousal violence)	0.167	NS	NA	NA
Women's decision (% of women who are involved in decision making)	0.107	NS	−0.168	NS
Community-level variables				
Sex ratio (number of women per 1000 men)	−0.133	NS	NA	NA
Crime against women (% contribution to all India total)	NA	NA	0.525	NS
Adjusted R^2	0.519	0.177		

[a] 1 Indian Rupee is approximately equivalent to US$0.02.
NS: not significant; NA: Not applicable. This variable is not included in multivariate regression analysis, as these variables have not met the selection criterion of a minimum *P* value of 0.25 in univariate regression analysis.

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able to resist and protest against the atrocities which they have been faced with due to the patriarchal norms. Women who are regularly exposed to media and with more access to modern educational systems challenge conventional thinking and lifestyles. Employed women are more likely to be exposed to media, and their financial resources allow them to question traditional practices which make them a burden to their families and undermine their status.[42] Women may also value the enhanced status in their conjugal families when they are given better dowries. The influences of modernization and self-interest may be paradoxical.

Poverty has multiple impacts in terms of everyone's life. Dowry is considered as one of the means to get rid of poverty, within the same class structure. Violence against women in the family was considered a family problem and never acknowledged as a crime or as a public health problem. It is discussed that women's lack of autonomy could be a major determinant of victimisation due to domestic violence.[43] There are reports of violence against women whose dowries are deemed insufficient by their husbands or the husbands' families.[44] Some socio-economic characteristics like lower education and lower family income of women have a significant association with the occurrence of domestic violence, and harassment by in-laws has been reported in some cases.[45] The third National Family Health Survey reported that more than 33% of women in India have been physically mistreated by their husbands or other family members.[27] Dowry related issues could be one of the reasons for this mistreatment.

Thus, this study attempted to identify some determinants of dowry-related deaths and suicides. However, before discussing the implications of the findings, it is important to note the limitations, as usual to this type of data and analysis. Like any analysis based on the information collected by different agencies under different governments, there might be reporting bias across the states in reporting of deaths. Also, the yearly increases of dowry deaths and suicides may partially be due to improved reporting practices. Another limitation is data of cross-sectional type from different sources, which do not allow us to establish cause-and-effect relationships. The associations observed during regression analyses could be the function of some prior cause. Also, there is the possibility of multicollinearity in multiple regression models used. Theoretically one can assume an association between two independent variables included in the model. Though this will not affect the overall prediction efficacy of the model, this will affect the coefficients of the individual predictors. Despite these limitations, this analysis had methodological strengths including the use of nation-level data from well-established surveys and from fifteen varied states to generalise the results to the country.

Reports reveal that the rates of dowry deaths and related crimes have been rising. Studies have been addressed in an attempt to try to understand the factors that make young women vulnerable to dowry-related violence. Dowry death might simply be a result of several factors including marriage patterns (especially hypergamy), the economic dependency of women and cultural norms that make the state's enacting agencies, such as the police and judiciary, hesitant to intervene in cases of domestic violence.[46] Present day prevailing dowry and dowry deaths are related to the frustration of urban middle class people, unable to afford the status and lifestyle they desire on their own earnings.[47]

It is evident that the conflict usually starts early in the marriage. A new bride is harassed when she brings a small amount of dowry and is compelled to bring more dowry from her family. However, her in-laws often remain unsatisfied and react to the situation with an attempt on the bride's life.[48] In order to curb this evil practice, The Dowry Prohibition Act 1961 was enacted, and amended in 1984 and 1986. The Act has been already questioned in terms of its existing ambiguity by experts. Most marriage negotiations are done as a confidential matter. Apart from this, both givers and takers of dowry are subject to criminal liability and so even if the bride is harassed by her in laws this will rarely be reported. Even after the legal recognition of crime, sometimes the concerned legal authorities consider violence against women in the home a private family matter.[49] Apart from the attitude of these authorities, the transformative law, which is supposed to make social transformation, has not considered the social customs, beliefs and values that would resist enforcement of these laws.[50] Since transformative law like The Dowry Prohibition Act has its own social, cultural and legislative limitations, there should be action-oriented initiatives and a comprehensive approach from all concerned individuals and institutions against dowry. The recent initiative from a *gram panchayat* (village-level self-government) in Kerala is a notable one.[51] Nilambur Grama Panchayat in Malappuram District of Kerala is expected to emerge as the first dowry-free panchayat in India. The Panchayat kicked off the campaign of being a 'dowry-free village' and set a new standard by pledging to 'neither ask for nor give any dowry' and has started a unique project against dowry. The Panchayat made initiatives to promote dowry-free marriages and to rehabilitate the victims of dowry abuse.

This analysis warrants that more research is required to better understand many societal, socio-cultural and economic issues responsible for this problem. In addition, there should be interventions from the health and welfare sectors of the government. Judicial stakeholders are also needed to look into the existing legislative deficiencies to combat the culturally-rooted malpractice in a better way. Moreover, the mindset of people is important and there should be strategies addressing discrimination against girls and women and the social norms that make dowry and dowry-related domestic violence acceptable within society. In addition, the present study warrants the addressing of the health consequences of dowry and dowry-related violence by the public health system. As advocated by earlier researchers,[34] there should be a national injury surveillance system. Also, there should be reliable estimates of dowry-related homicides. Despite the lack of such a system and estimates, efforts should be made to assess the present situation of the problem from other sources such as NCRB,[26] and other methods such as the compilation of data of medical certification of cause of death in urban hospitals,[52] surveys of causes of death[53] and the sample registration systems of births and deaths.[54] This

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information can be used to design and implement countermeasures to prevent dowry-related violence and deaths. Research among survivors of such violence, and family and neighbours of victims will give some insight into circumstances and triggers of this violence. Also, this research will provide insight into healthcare seeking of these victims, difficulties in seeking health care and reporting to the police. This type of information is more likely to translate into some preventive interventions.

Authors' contributions: GRB compiled the data, analysed and interpreted the data and wrote the first draft. BVB developed the data analysis plan, interpreted the results and refined the manuscript. Both the authors read and approved the final version. GRB is guarantor of the paper.

Funding: None.

Conflicts of interest: None declared.

Ethical approval: Not required.

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